

BANARAS HINDU UNIVERSITY

B.A. (Hons.) Semester V Examination, 2022-23

Paper No. STA-401: Sample Surveys and Design of Experiments

Time: 3 Hours

Maximum Marks: 70

*Instructions: Answer **five** questions including Question 1 which is compulsory. All questions carry equal marks. Symbols have their usual meanings.*

Question 1. What is a sample survey? Discuss the basic principles of a sample survey. Write down its objectives and its advantages over the census. [14]

Question 2. A population consists of 5 units with respective values 25, 34, 32, 42, and 30. Write down all possible samples of size 2 using simple random sampling with replacement (SRSWR) and simple random sampling without replacement (SRSWOR) schemes. Estimate the population mean and variance using these samples. Give comments on the results. [14]

Question 3. Show that in SRSWOR, $\text{Cov}(\bar{y}, \bar{x}) = \frac{1-f}{n} S_{xy}$, where $S_{xy} = \frac{1}{N-1} \sum_{i=1}^N (X_i - \bar{X})(Y_i - \bar{Y})$ and $f = \frac{n}{N}$ is the sampling fraction. Also write down the formula for the unbiased estimator of $\text{Cov}(\bar{y}, \bar{x})$ under both SRSWR and SRSWOR schemes. [14]

Question 4. Explain stratified random sampling with an example and discuss the allocation problem. Show that:

$$V(\bar{y}_{\text{srs}}) \geq V(\bar{y}_{\text{st}})_{\text{prop}} \geq V(\bar{y}_{\text{st}})_{\text{Ney}}$$

where the notations have their usual meanings. [14]

Question 5. Discuss systematic and cluster sampling designs. Derive the estimator of the population mean along with its standard error in the case of systematic sampling. State the conditions under which systematic sampling provides a more efficient estimate than stratified random sampling. [14]

Question 6. Discuss ratio and regression estimation methods. Do these methods provide unbiased estimators of the population mean? Obtain the mean squared error (MSE) of the population mean estimator in both cases. Compare the results with that of simple random sampling. [14]

Question 7. Discuss the significance of the analysis of variance (ANOVA). State the basic assumptions underlying the analysis of variance. Describe the two-way classification layout, parameter estimation, and the ANOVA table. [14]

Question 8. What do you mean by the design of experiments? State its basic principles. Discuss the Latin Square Design (LSD) along with its mathematical model, parameter estimation, and the ANOVA table. [14]